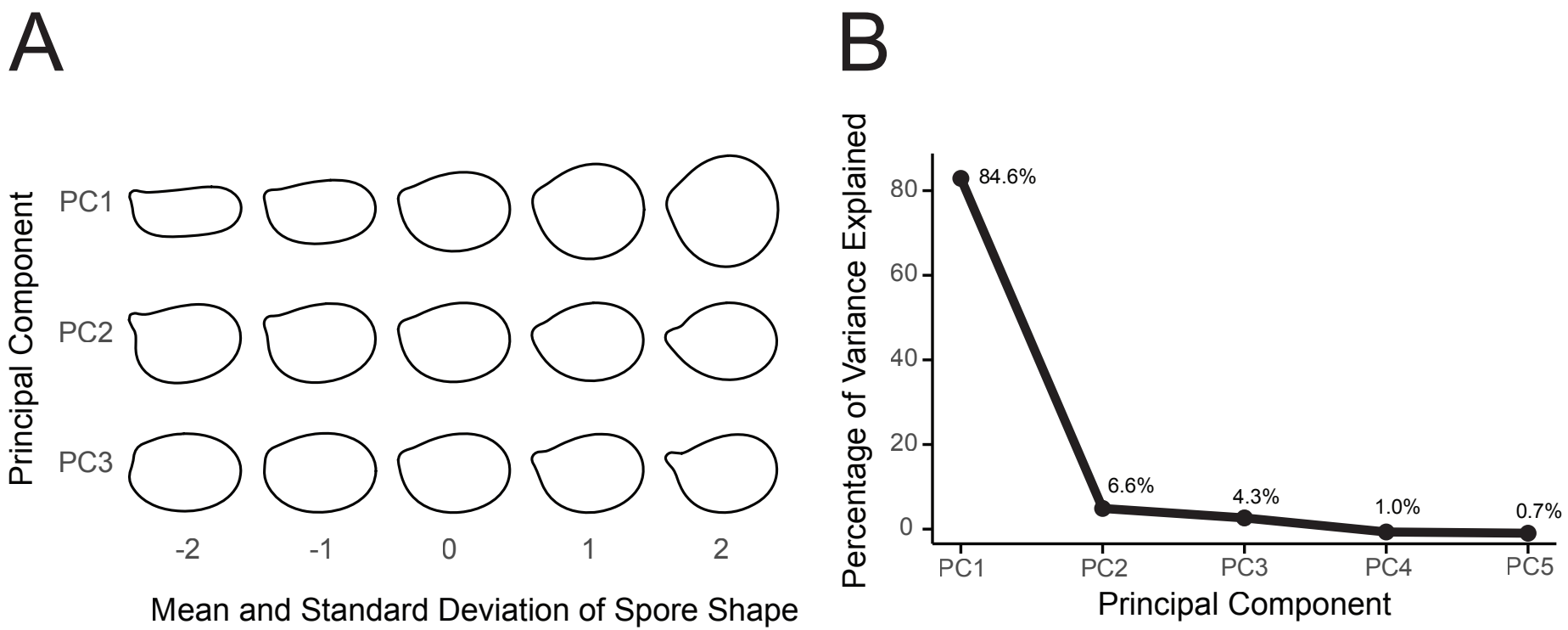


Figure. 3 Fourier transform and principal component analysis results. (a) Variation in *Amanita* spore shape along the axes of the top 3 principal components. Principal component 1 (PC1) is analogous to the “Elongate-Globose” spectrum qualitatively described by taxonomists. Principal component 2 (PC2) represents the placement of the hilar appendix. Principal component 3 (PC3) represents the prominence of the hilar appendix. (b) Variance explained by first 5 principal components. Principal component 1 explains the vast majority of variation.